

Problem 117. Euclid placed 8 points on the plane. Then he writes down all the areas of the triangles (a total of 56 numbers) constructed by these points.

Prove that he can always assign positive or negative signs to these numbers such that the sum of these numbers becomes zero.

Solution. We first prove this for any 4 points A, B, C, D on the plane. If $ABCD$ form a convex quadrilateral, including the limiting case when some 3 or all points are collinear, we have $[ABC] + [ADC] - [ABD] - [CBD] = 0$. If D lies inside $\triangle ABC$, we have $[ABC] - [CAD] - [ABD] - [BCD] = 0$.

Now, it suffices to partition the 56 triangles into 14 groups of 4 triangles, so that each group consists of all triangles formed by a quadrilateral. A partition is as follows (where the numbers mean the labels of the points):

$$\begin{aligned} &1234, 1256, 1278, 1357, 1368, 1458, 1467, \\ &2358, 2367, 2457, 2468, 3456, 3478, 5678. \end{aligned}$$

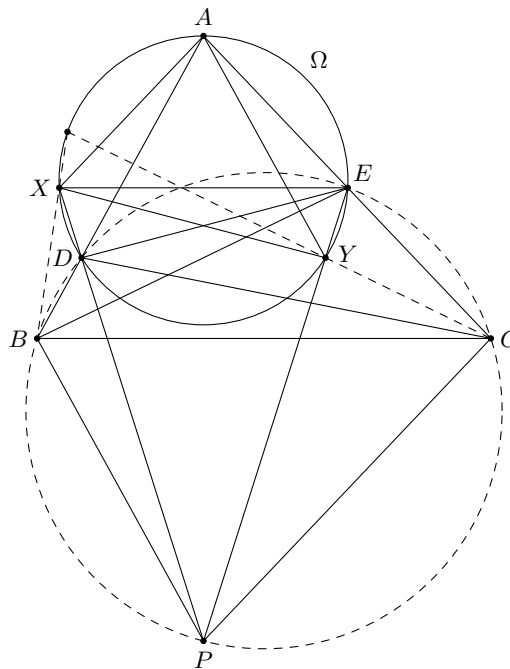
□

Problem 118. In $\triangle ABC$, with $AB < AC$, let

- D and E be points on the sides AB and AC , respectively, with $D, E \neq A, B, C$, such that $CA = CD$ and $BA = BE$;
- Ω be the circumcircle of $\triangle ADE$;
- P be reflection of A about BC ;
- $X = PD \cap \Omega$ and $Y = PE \cap \Omega$ with $X \neq D$ and $Y \neq E$.

Prove that the two lines BX and CY intersect at a point lying on Ω .

Solution.



Since $\angle BPC = \angle CAB = \angle ADC = \angle BDC$, the points B, P, C, D are concyclic. Similarly, B, P, C, E are concyclic. This means B, P, C, E, D are concyclic.

Next, we have $\angle AXY = \angle AEY = \angle CEY = \angle CEP = \angle CBP = \angle ABC$. Similarly, we have $\angle AYX = \angle ACB$. It follows that $\triangle AXY$ and $\triangle ABC$ are directly similar. By properties of spiral similarity, the intersection of BX and CY lie on (AXY) as desired.

□

Problem 119. Let $S = \{1, 2, \dots, 999\}$. Suppose a function $f: S \rightarrow S$ satisfies the following condition:

$$f^{n+f(n)+1}(n) = f^{nf(n)}(n) = n \quad \text{for every } n \in S.$$

Prove that there exists an element $a \in S$ which satisfies the condition $f(a) = a$.

Here $f^k(n)$ means the iteration of the function $f(n)$ k -times, i.e., $f^2(n) = f(f(n))$, $f^3(n) = f(f(f(n)))$, etc.

Solution. Since $n + f(n) + 1 > 1$, there exists a smallest positive integer c_n satisfying $f^{c_n}(n) = n$. It is easy to show that $f^i(n) = f^j(n)$ if and only if $i \equiv j \pmod{c_n}$. Equivalently, the numbers $n, f(n), f^2(n), \dots, f^{c_n-1}(n)$ are pairwise distinct, and the pattern repeats every c_n terms. This means every element n in S belongs to a cycle of length c_n , such that $f^i(n)$ takes up exactly c_n different possible values. Obviously, each element belongs to exactly one such cycle.

Since there are $999 \equiv 1 \pmod{2}$ elements in S , there must be a cycle with an odd length. Consider any such cycle $n, f(n), f^2(n), \dots, f^{c-1}(n)$ with odd length c . Since c is the period and $f^{n+f(n)+1}(n) = n$, we have $c \mid n + f(n) + 1$. Similarly, we have $c \mid nf(n)$. From $c \mid n + f(n) + 1$, we obtain $f(n) \equiv -(n+1) \pmod{c}$. Replacing n by $f(n)$, and noting that the length c is unchanged, we have $f^2(n) \equiv -(f(n) + 1) \equiv n \pmod{c}$. Inductively, $f^i(n) \equiv n \pmod{c}$ for even i and $f^i(n) \equiv -(n+1) \pmod{c}$ for odd i . In particular, $n = f^c(n) \equiv -(n+1) \pmod{c}$ since c is odd. This implies $c \mid 2n + 1$.

Recall that $c \mid n + f(n) + 1$ and $c \mid nf(n)$. Eliminating $f(n)$, we obtain $c \mid n(-n-1)$, i.e. $c \mid n(n+1)$. However, it is clear that $(n, 2n+1) = (n, 1) = 1$ and $(n+1, 2n+1) = (n+1, 1) = 1$. Therefore, from $c \mid (2n+1, n(n+1))$, we obtain $c = 1$. This means exactly $f(n) = n$. □

Problem 120. The sequence $a_1, a_2, \dots$ satisfies the equality

$$\sum_{k=1}^n a_{\lfloor \frac{n}{k} \rfloor} = n^{10} \quad \text{for all positive integers } n.$$

Given an arbitrary positive integer c , prove that

$$\frac{c^{a_n} - c^{a_{n-1}}}{n}$$

is an integer for each $n > 1$.

Here $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

Solution. We have $\sum_{k=1}^n a_{\lfloor \frac{n}{k} \rfloor} = n^{10}$ and $\sum_{k=1}^{n-1} a_{\lfloor \frac{n-1}{k} \rfloor} = (n-1)^{10}$ for $n > 1$. Note that $\lfloor \frac{n}{k} \rfloor = \lfloor \frac{n-1}{k} \rfloor$ unless $k \mid n$, in which case $\lfloor \frac{n}{k} \rfloor = \lfloor \frac{n-1}{k} \rfloor + 1$. Therefore, taking the difference of the two equations, and by defining $b_1 = a_1 = 1$ and $b_m = a_m - a_{m-1}$ for $m > 1$, we obtain $\sum_{\substack{k=1 \\ k \mid n}}^{n-1} b_{\frac{n}{k}} + a_1 = n^{10} - (n-1)^{10}$. This means $\sum_{k \mid n} b_k = n^{10} - (n-1)^{10}$. Note that it also holds for $n = 1$. Using the Möbius inversion formula, we get

$$b_n = \sum_{k \mid n} \mu(k) \left(\left(\frac{n}{k} \right)^{10} - \left(\frac{n}{k} - 1 \right)^{10} \right) = \sum_{r=0}^9 \binom{10}{r} (-1)^{r+1} \sum_{k \mid n} \mu(k) \left(\frac{n}{k} \right)^r.$$

Note that $\sum_{k \mid n} \mu(k) \left(\frac{n}{k} \right)^r = n^r \left(1 - \frac{1}{p_1^r} \right) \left(1 - \frac{1}{p_2^r} \right) \cdots \left(1 - \frac{1}{p_s^r} \right)$ where $p_1, p_2, \dots, p_s$ are all the distinct prime divisors of n (consider the expansion of the expression on the right and the definition of the Möbius function). For $r > 0$, this is further equal to

$$\varphi(n) n^{r-1} \left(1 + \frac{1}{p_1} + \cdots + \frac{1}{p_1^{r-1}} \right) \cdots \left(1 + \frac{1}{p_s} + \cdots + \frac{1}{p_s^{r-1}} \right).$$

Therefore, for $n > 1$, we have

$$\frac{b_n}{\varphi(n)} = \sum_{r=1}^9 \binom{10}{r} (-1)^{r+1} n^{r-1} \prod_{p \mid n} \left(1 + \frac{1}{p} + \cdots + \frac{1}{p^{r-1}} \right).$$

It is clear that the right-hand side is an integer. Therefore, $\varphi(n) \mid b_n$ for any $n > 1$. This also holds for $n = 1$ by definition. Also, one can use this to show that $b_n > 0$. The base cases $b_1 = 1, b_2 = 2^{10} - 2, b_3 = 3^{10} - 2^{10} - 1$ and $b_4 = 4^{10} - 3^{10} - 2^{10} + 1$ can be verified directly. For $n \geq 5$, we find that

$$\begin{aligned} \frac{b_n}{\varphi(n)} &> (10n - 45)n^7 \prod_{p \mid n} \left(1 + \frac{1}{p} + \cdots + \frac{1}{p^7} \right) + (120n - 210)n^5 \prod_{p \mid n} \left(1 + \frac{1}{p} + \cdots + \frac{1}{p^5} \right) \\ &\quad + (252n - 210)n^3 \prod_{p \mid n} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} \right) + (120n - 45)n \prod_{p \mid n} \left(1 + \frac{1}{p} \right) + 10 > 0. \end{aligned}$$

From $b_n > 0$ we obtain $a_n \geq n$.

Now, for any fixed c and n , write $n = n_1 n_2$ where n_1 is composed of the prime divisors of c and $(n_2, c) = 1$. Then it is easy to see that $n_1 \mid c^{n_1-1} \mid c^{n-1} \mid c^{a_{n-1}}$. Also, we have $n_2 \mid c^{\varphi(n_2)} - 1 \mid c^{\varphi(n)} - 1 \mid c^{b_n} - 1$. Therefore, $n = n_1 n_2 \mid c^{a_{n-1}} (c^{b_n} - 1) = c^{a_n} - c^{a_{n-1}}$. □